

- 3 (a) (i) The kinetic theory for an ideal gas of volume V at pressure p leads to the equation

$$pV = \frac{1}{3}Nm\langle c^2 \rangle,$$

where the other symbols refer to their usual meanings.

Use the equation of state for an ideal gas to show that the average translational kinetic energy E_K of a molecule of ideal gas is given by

$$E_K = \frac{3}{2}kT.$$

$$pV = NkT$$

$$\frac{1}{3}Nm\langle c^2 \rangle = NkT$$

$$\frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT \quad [\text{B1}] \text{ for } pV = NkT \text{ leading to } \frac{1}{2}m\langle c^2 \rangle \text{ seen}$$

[1]

- (ii) One helium atom has a mass of 6.68×10^{-27} kg. Helium may be considered as an ideal gas.

Show that the total kinetic energy of the helium atoms in 1.00 mol of helium gas at 25 °C is 3720 J.

$$\begin{aligned} E_{K, \text{ total}} &= \frac{3}{2}NkT \\ &= \frac{3}{2}(6.02 \times 10^{23})(1.38 \times 10^{-23})(25 + 273.15) \quad [\text{C1}] \\ &= 3715 \\ &= 3720 \text{ J (shown)} \quad [\text{A0}] \end{aligned}$$

[1]

- (iii) State the value of the internal energy of 1.00 mol of helium gas at 25 °C. Explain your answer.

Internal energy is the sum of the random distribution of the microscopic kinetic energy (KE) and microscopic potential energy (PE) of the gas molecules [B1]

For an ideal gas, there are no intermolecular forces between molecules so PE = 0, so internal energy = KE = 3720 J [B1] [2]

- (iv) The helium gas is gradually cooled from 25 °C to –150 °C at which the internal energy is 1540 J.

On Fig. 3.1, plot points and draw a line to show the variation with temperature θ of the internal energy U of the helium gas.

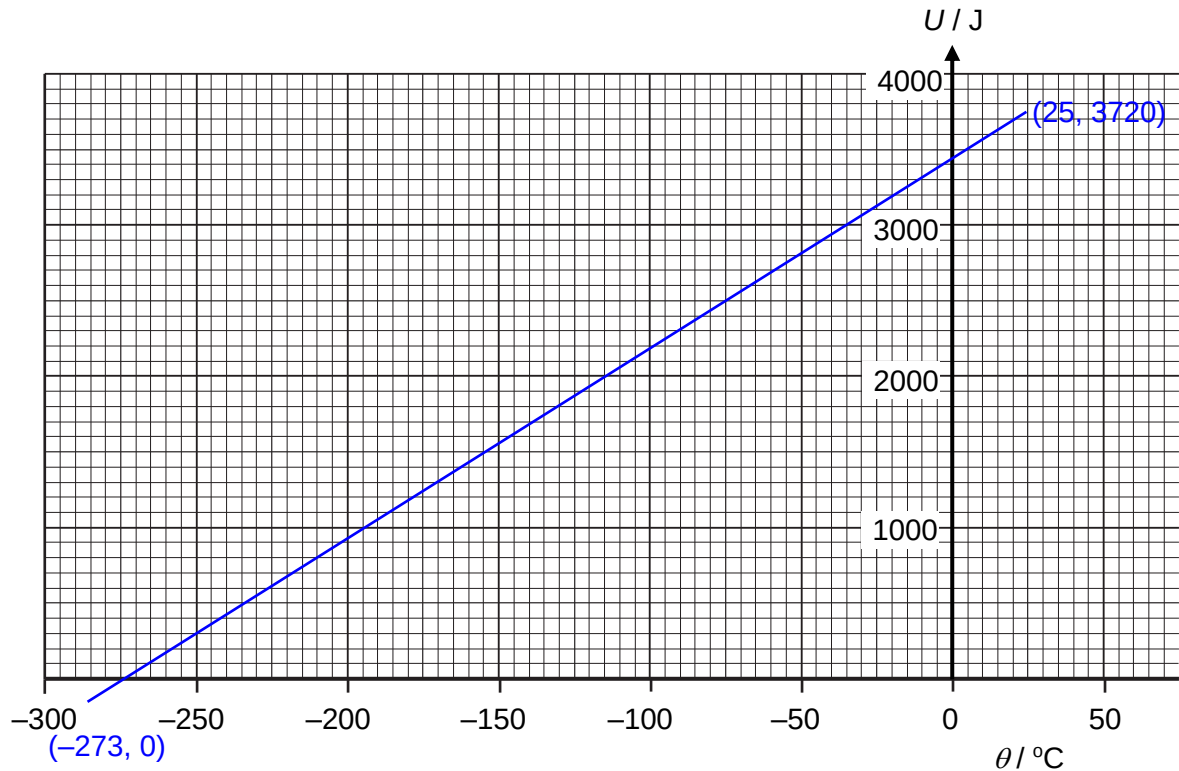


Fig. 3.1

[1]

- (v) Explain how your graph leads to the idea of an absolute zero of temperature.

The graph shows a linear relationship between internal energy and temperature so
extrapolate the graph to 0 internal energy to reach a temperature of –273.15 °C [B1]

[1]

- (b) Gases like hydrogen and helium are found mainly in stars. These gases are at a very high pressure.

Use the assumptions of the kinetic theory of gases to suggest why, in practice, the gas found in stars is unlikely to behave as an ideal gas.

At very high pressure, gas molecules are very close to each other [B1]
EITHER intermolecular forces are not negligible OR volume of molecules are not negligible
compared with the gas volume [B1]

[2]

[Total: 8]

4. A tube closed at one end has a constant area of cross section A . Some lead shots are placed